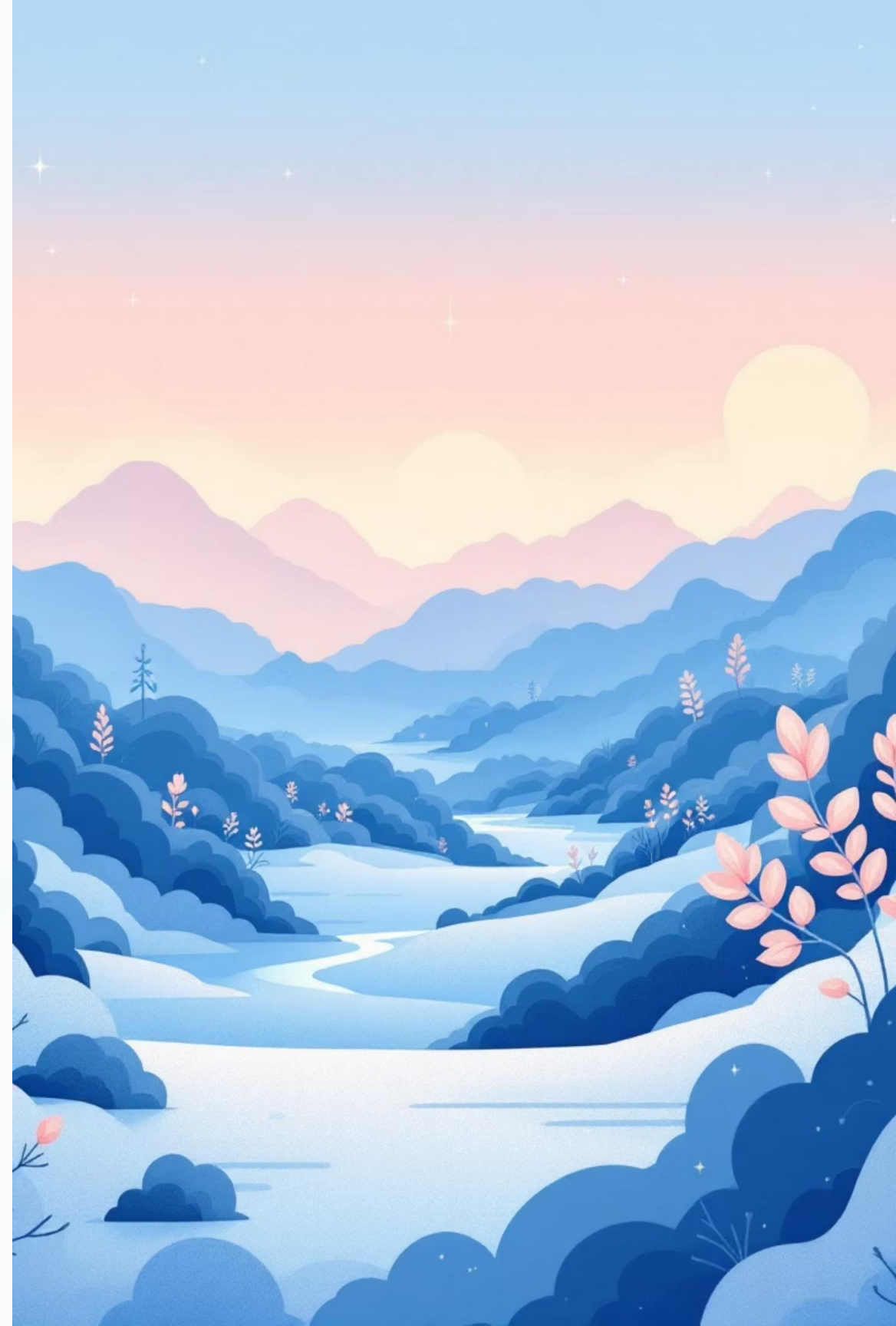


Enhanced YouTube Comment Analysis using Three SVM-Based NLP Models and a Streamlit Dashboard



Introduction

YouTube comments, a dynamic and diverse space, often present challenges for content creators, educators, brands, and communities. The prevalence of toxicity, spam, and negative sentiment can detract from constructive discourse and impact user experience. An automated, AI-driven moderation system is therefore crucial to maintain a healthy and engaging online environment.

The Problem: Unmoderated Comments

YouTube's vast comment sections are frequently plagued by:

- Toxic language
- Spam and irrelevant content
- Overwhelmingly negative sentiment

This degrades community health and engagement.

The Solution: AI Moderation

An automated AI-based moderation system can provide:

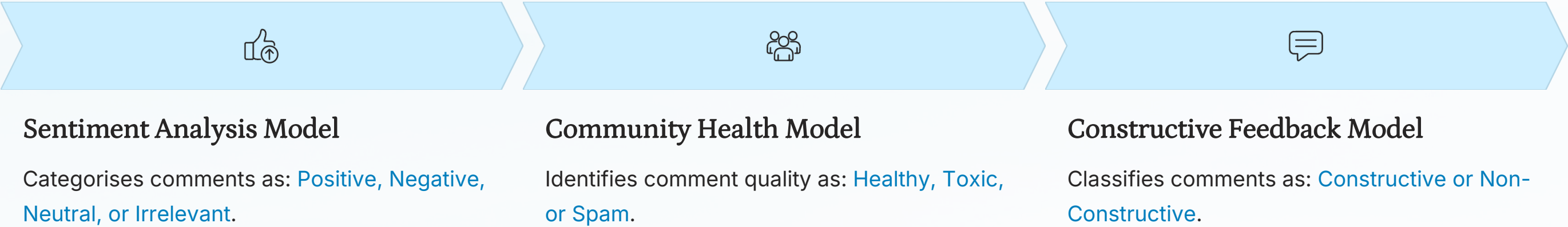
- Real-time content filtering
- Improved user experience
- Protection for creators and brands

Fostering positive online interactions.



Proposed System

Our system analyses YouTube comments through three distinct Support Vector Machine (SVM) models, each trained for a specific moderation aspect. These models work in conjunction to provide a comprehensive evaluation of comment quality, with all results seamlessly integrated into an interactive Streamlit-based dashboard.



The outputs of these three models are then unified within a [Streamlit dashboard](#), offering a holistic view for content moderation.

Dataset Description

Each SVM model was trained on a specialised dataset, carefully selected and preprocessed to ensure optimal performance for its specific task. These diverse datasets cover a wide range of comment types and moderation challenges.



Sentiment Model Dataset

Leveraged the [Twitter Training Dataset](#), comprising millions of tweets annotated for sentiment. This dataset allowed for robust training on nuanced emotional expressions.



Community Health Dataset

A combination of the [YouTube Spam Collection](#) and the [Jigsaw Toxic Comment Dataset](#). This dual approach provided comprehensive coverage for identifying spam, toxicity, and other harmful content.



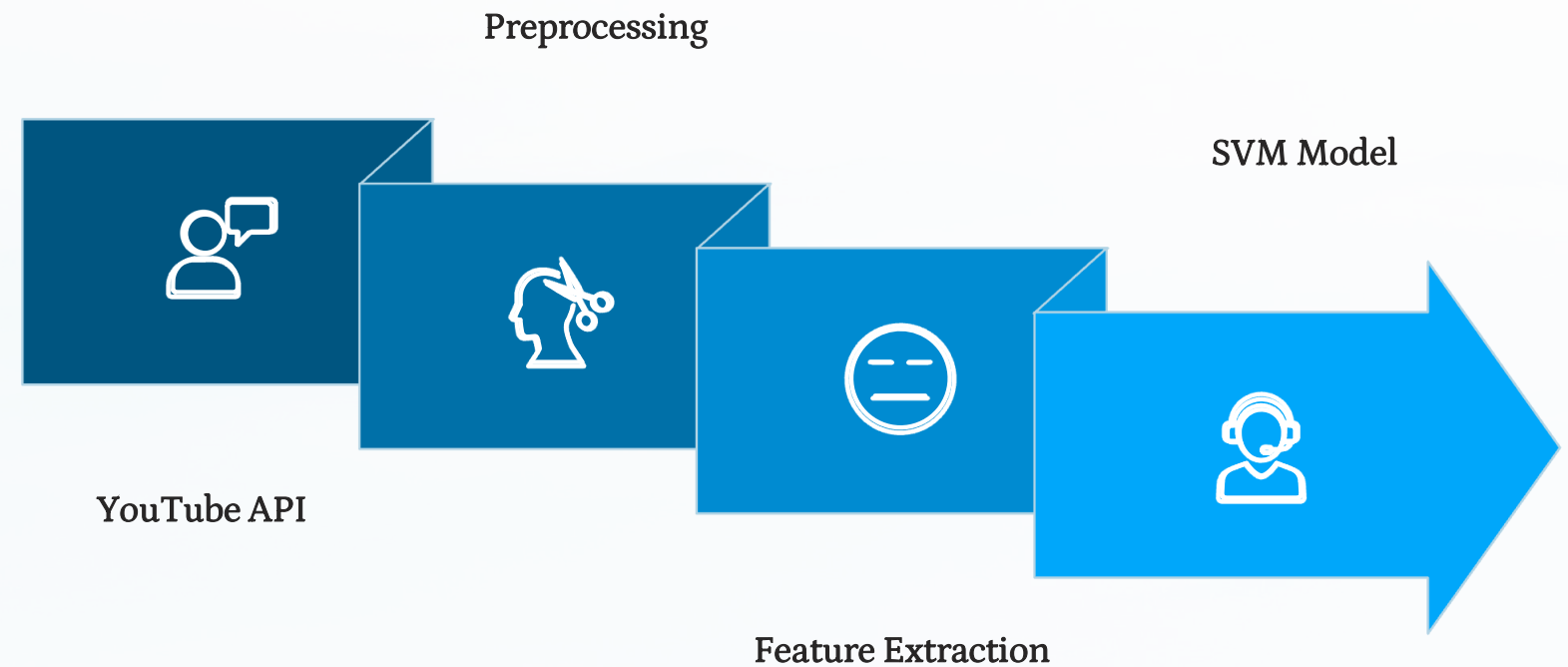
Constructive Feedback Dataset

Utilised the [C3 Constructive Comments Dataset](#), specifically designed for classifying constructive discourse. This enabled the model to discern valuable feedback from general commentary.

All datasets underwent rigorous preprocessing, including tokenisation, stop-word removal, and stemming, to prepare the text for feature extraction and model training. The label categories for each dataset were carefully mapped to align with the desired output of each model.

Methodology

The overall system follows a structured pipeline, from data acquisition to dashboard visualisation. This ensures efficient processing and accurate analysis of YouTube comments.



This sequential approach allows for modular development and ensures that each stage of the analysis is optimised for performance and accuracy.

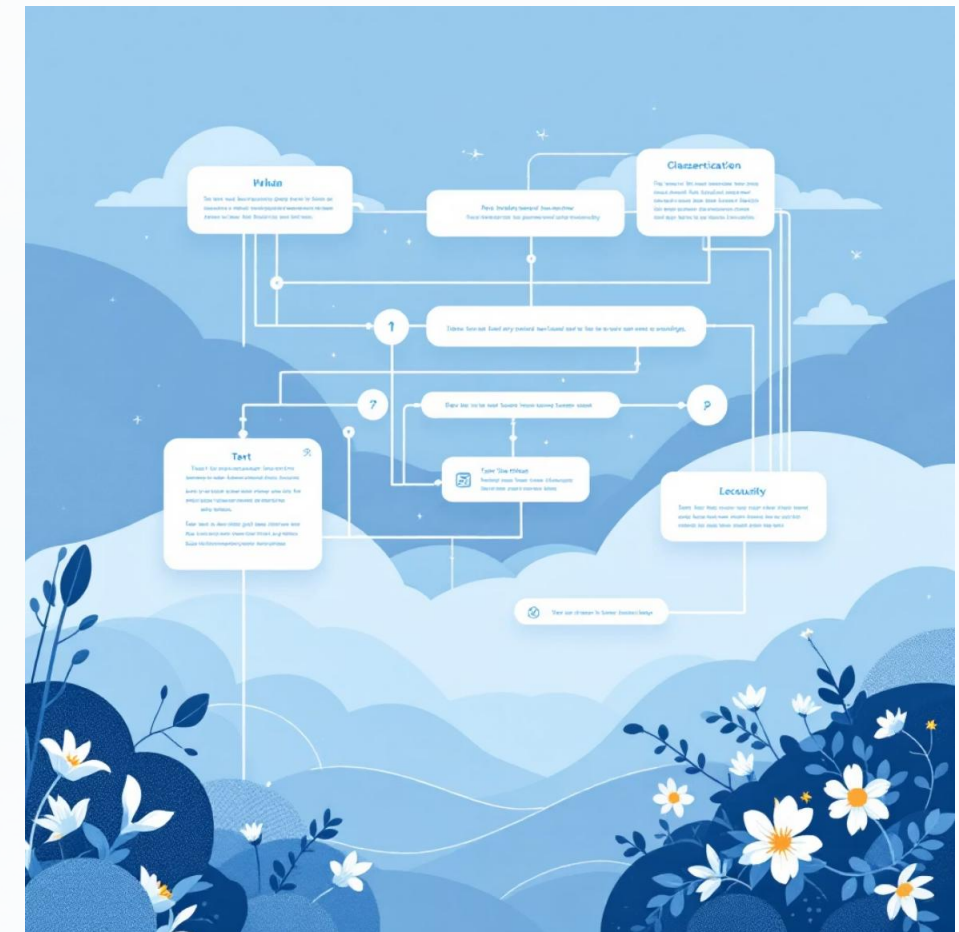
Model Architecture

Our moderation system employs a consistent yet adaptable model architecture across all three SVM-based classifiers, ensuring both efficiency and specialised performance for each task.

Core Components

- All three models use **TF-IDF Vectorisation** for converting text data into numerical features, capturing the importance of words within comments.
- The Constructive Feedback model additionally incorporates **comment length as a feature**, applying **MinMaxScaler** and **hstack()** for integration, enhancing its ability to identify valuable contributions.
- A **Linear SVC (Support Vector Classifier)** with **class_weight='balanced'** is used across all models to effectively handle imbalanced datasets.

This design choice for separate individual ML training pipelines, rather than a single multi-label network, allows for greater fine-tuning and avoids potential issues from conflicting feature importance or label interdependence across diverse moderation tasks.



Training Pipeline

A robust training pipeline was implemented for each model, focusing on rigorous evaluation and optimisation to achieve high accuracy and generalisability. This iterative process ensures that each model is finely tuned for its specific moderation task.



Data Splitting

Performed [train/test splits](#) to evaluate model performance on unseen data, preventing overfitting.



Cross-Validation & Hyperparameter Tuning

Employed [cross-validation](#) techniques and [hyperparameter tuning](#) to optimise model parameters for peak performance.



Bias and Variance Analysis

Utilised [learning curves](#) to diagnose and address issues of bias and variance, ensuring a balanced model.



Performance Metrics

Generated [confusion matrices](#) for detailed classification performance analysis, including precision, recall, and F1-score.



Model Export

Final trained models were exported as [.joblib files](#) for efficient deployment and integration into the dashboard.

Implementation

The entire system is engineered for seamless integration and user-friendly interaction, from the backend machine learning models to the frontend dashboard and live data fetching.



Backend ML Models

All trained models are stored as `sentiment_model.joblib`, `community_model.joblib`, and `constructive_model.joblib` for easy access and deployment.



Frontend Dashboard

The interactive dashboard is built using `app.py` and implemented with [Streamlit](#), providing an intuitive interface for users.



Live Data Extraction

Comments are extracted live from YouTube using the [YouTube Data API](#), ensuring real-time analysis.



Dynamic Visualisations

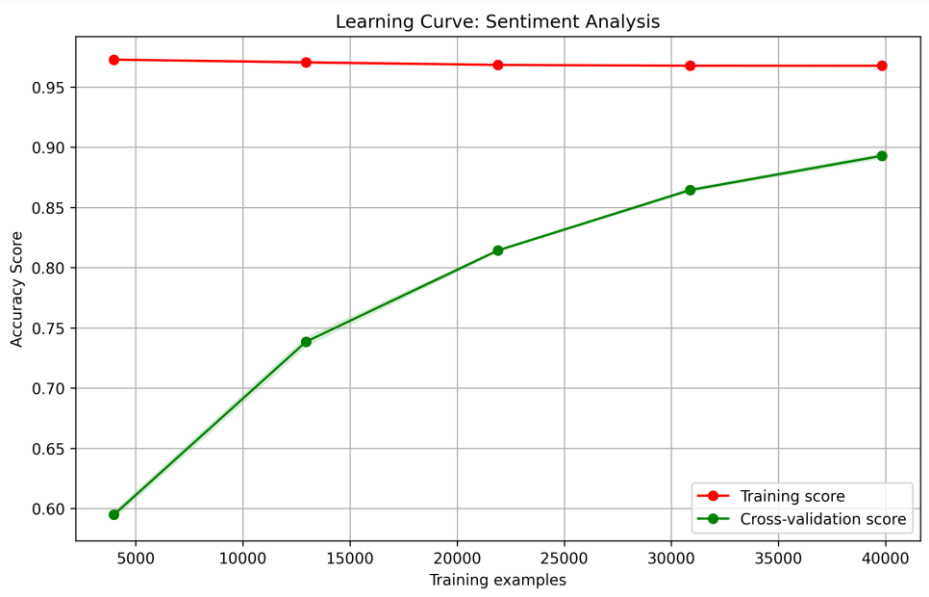
Data is presented with [Altair Charts and DataFrames](#), offering dynamic and insightful visualisations of comment analysis.

This architecture ensures a powerful and responsive tool for YouTube comment moderation.

Results

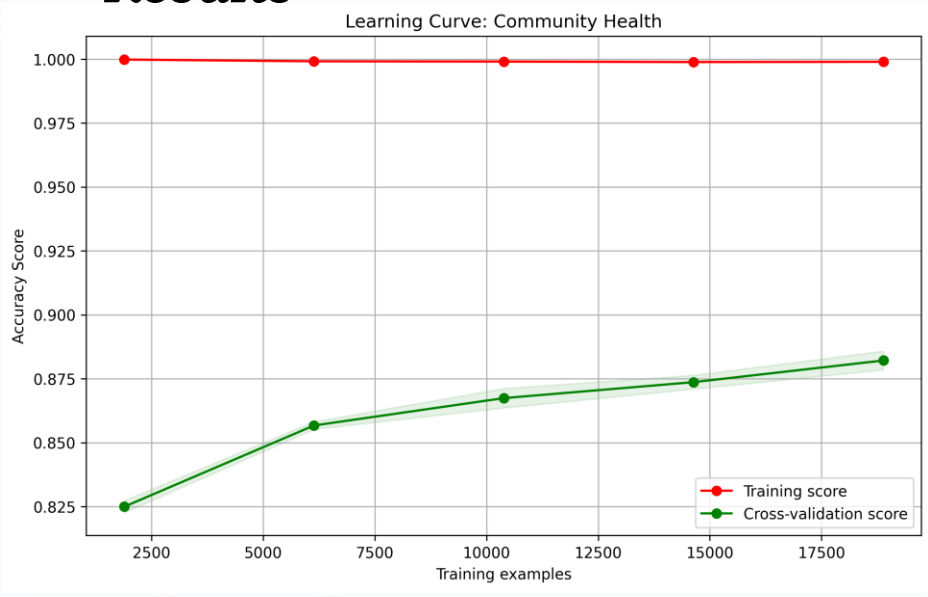
The performance of each model was rigorously evaluated using key metrics, learning curves. These visualisations provide a clear understanding of each model's strengths and areas for potential improvement.

Sentiment Model Results



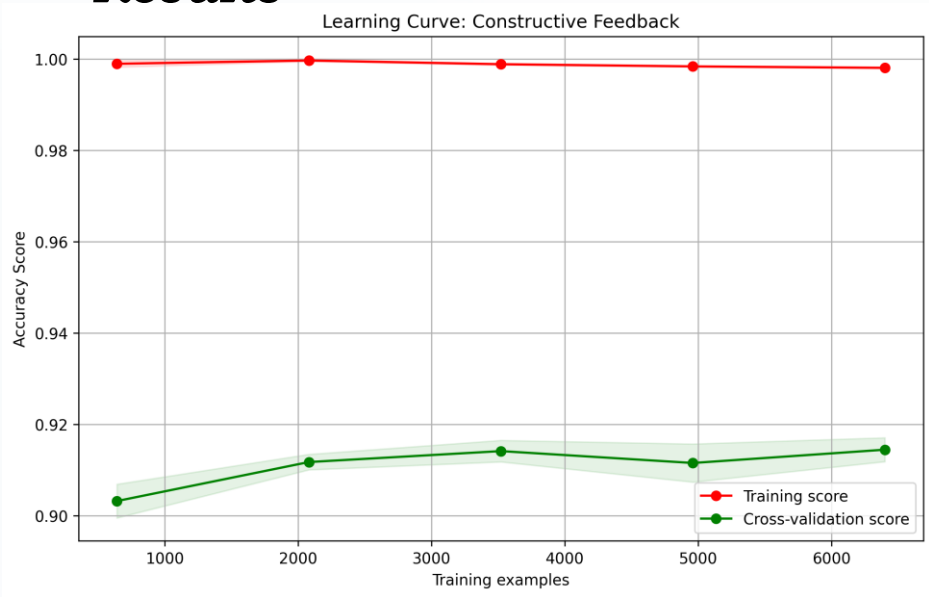
Accuracy: 0.9213

Community Health Model Results



Accuracy: 0.8967

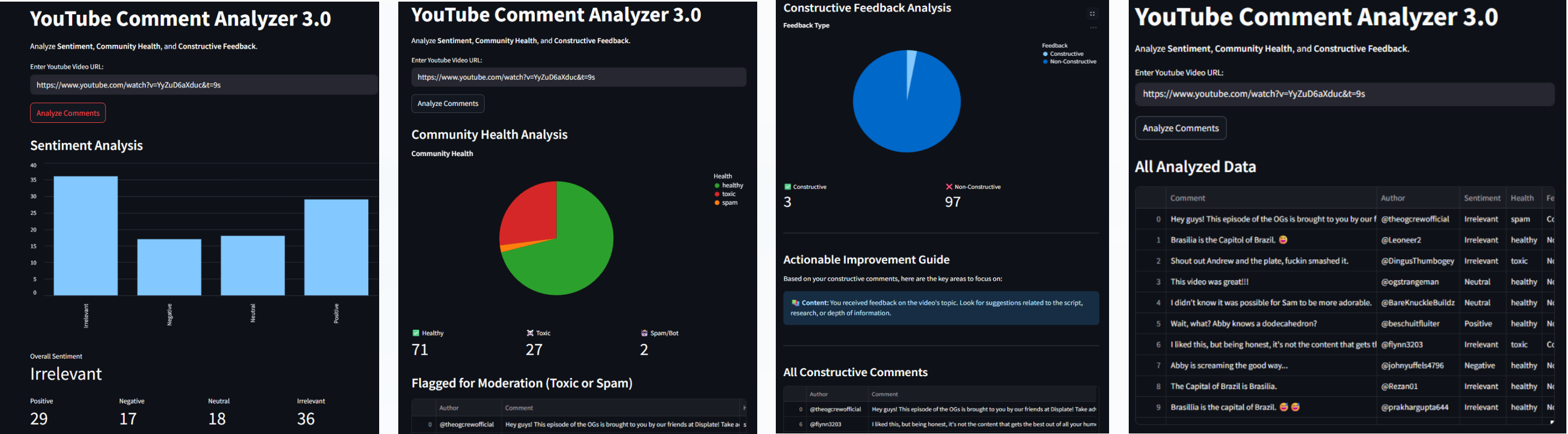
Constructive Feedback Model Results



Accuracy: 0.9167

Dashboard Snapshots

Our Streamlit-based dashboard provides an intuitive and comprehensive interface for visualising the results of the comment analysis. Here are some key views of the dashboard.



These visualisations allow users to quickly grasp the overall health of their comment sections and identify actionable insights for moderation.